

Tools for Identifying Malicious Activity (CS0-003)

Introduction

Overview

Learning Outcomes:

In this module, you will complete the following exercises:

- Exercise 1 – Monitoring Network Activity
- Exercise 2 – Using a Sandbox for Analyzing Malicious Files
- Exercise 3 – Validating Domain Names and IP Addresses

After completing this module, you should be able to:

- Customize the Wireshark Application
- Capture Network Traffic using the Wireshark Application
- Capture Network Traffic using Tcpdump
- Analyze Malicious Code
- Use WHOIS to Validate Domain Names
- Use AbuseIPDB to Validate Domain Names
- Use the Virustotal Website to Verify Links

Exam Objectives:

The following exam objectives are covered in this module:

1.3 Given a scenario, use appropriate tools or techniques to determine malicious activity.

- Tools

Objective

Welcome to the **Tools for Identifying Malicious Activity** practice lab. In this module, you will be provided with the instructions and devices needed to develop your hands-on skills.

Exercise 1 – Monitoring Network Activity

As a Cyber Security Analyst, several tools are available that can be used to monitor and analyze network traffic. Analyzing network traffic is essential to detect any malicious activity on the network.

In this exercise, different network tools will be used to demonstrate how network traffic can be monitored and analyzed for malicious activity.

Learning Outcomes

After completing this exercise, you should be able to:

- Customize the Wireshark Application
- Capture Network Traffic using the Wireshark Application
- Capture Network Traffic using Tcpdump

Your Devices

You will be using the following devices in this lab. Please power these on now.

- **ACIDC01** – Windows Server 2022 – Domain Controller
- **ACIDM01** – Windows Server 2022 – Domain Member Server
- **ACIWIN11** – Windows 11 – Domain Member Workstation
- **ACIALMA** – ALMA Linux Standalone Workstation
- **ACIKALI** – Kali Linux Standalone Workstation

Task 1 – Customizing the Wireshark Application

The Wireshark application's view can be customized according to the needs of a Cyber Security Analyst. The columns can be changed accordingly depending on the type of traffic that needs to be monitored.

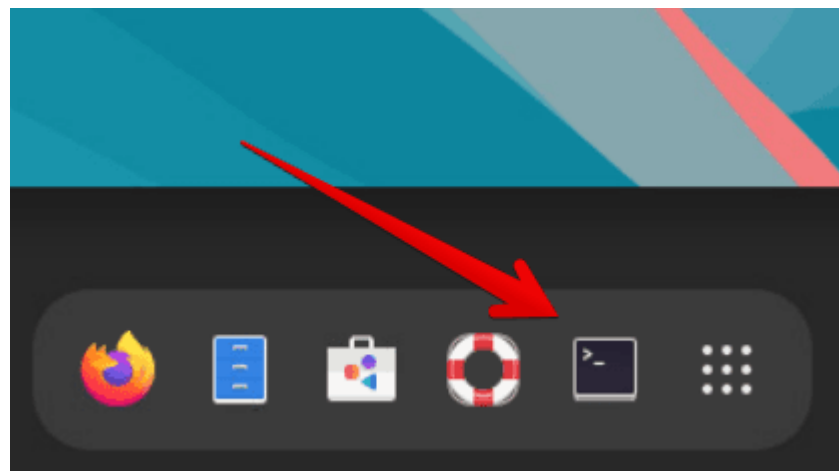
In this task, the Wireshark application will be customized to view different types of captured network traffic.

1

Click the **Start** button.



- 2 **Connect** to **ACIALMA**, and **select** **Terminal** from the **Activities** menu.



- 3 In the **aciadmin@localhost:~** window, **type** the **following** and **press** **Enter**:

```
sudo wireshark
```

When prompted, **enter** the **following** password:

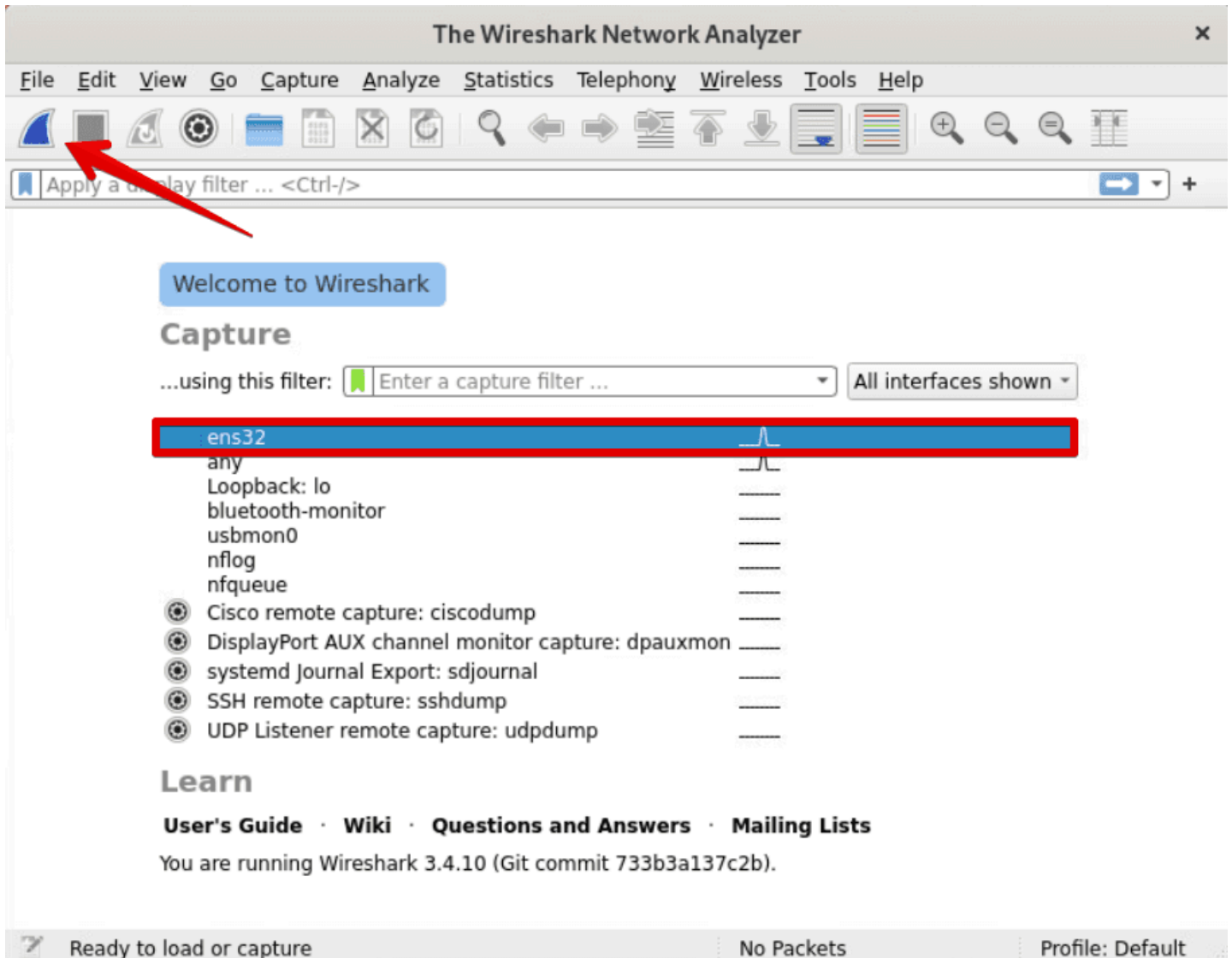
```
Passw0rd
```

Press **Enter**.

```
[aciadmin@localhost ~]$ sudo wireshark
[sudo] password for aciadmin: Passw0rd
```

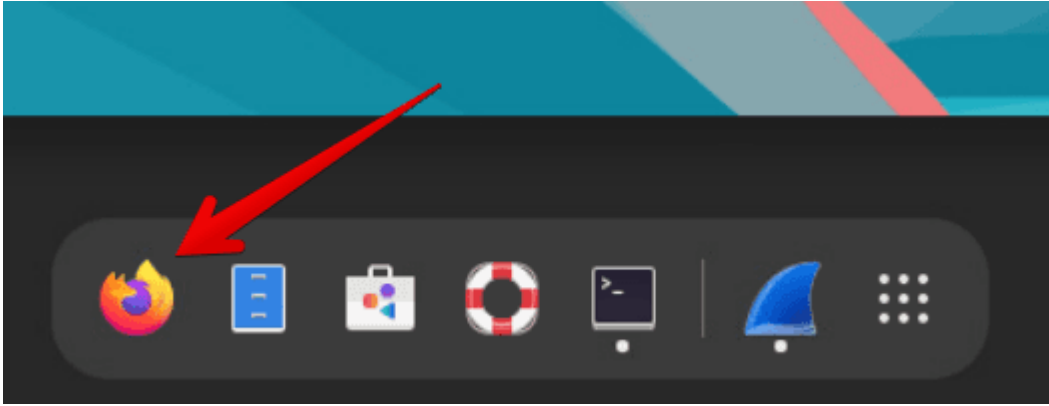
4

In the **Wireshark Network Analyzer** window, **ensure** **ens32** is selected and **click** **Start Capturing Packets**.



5

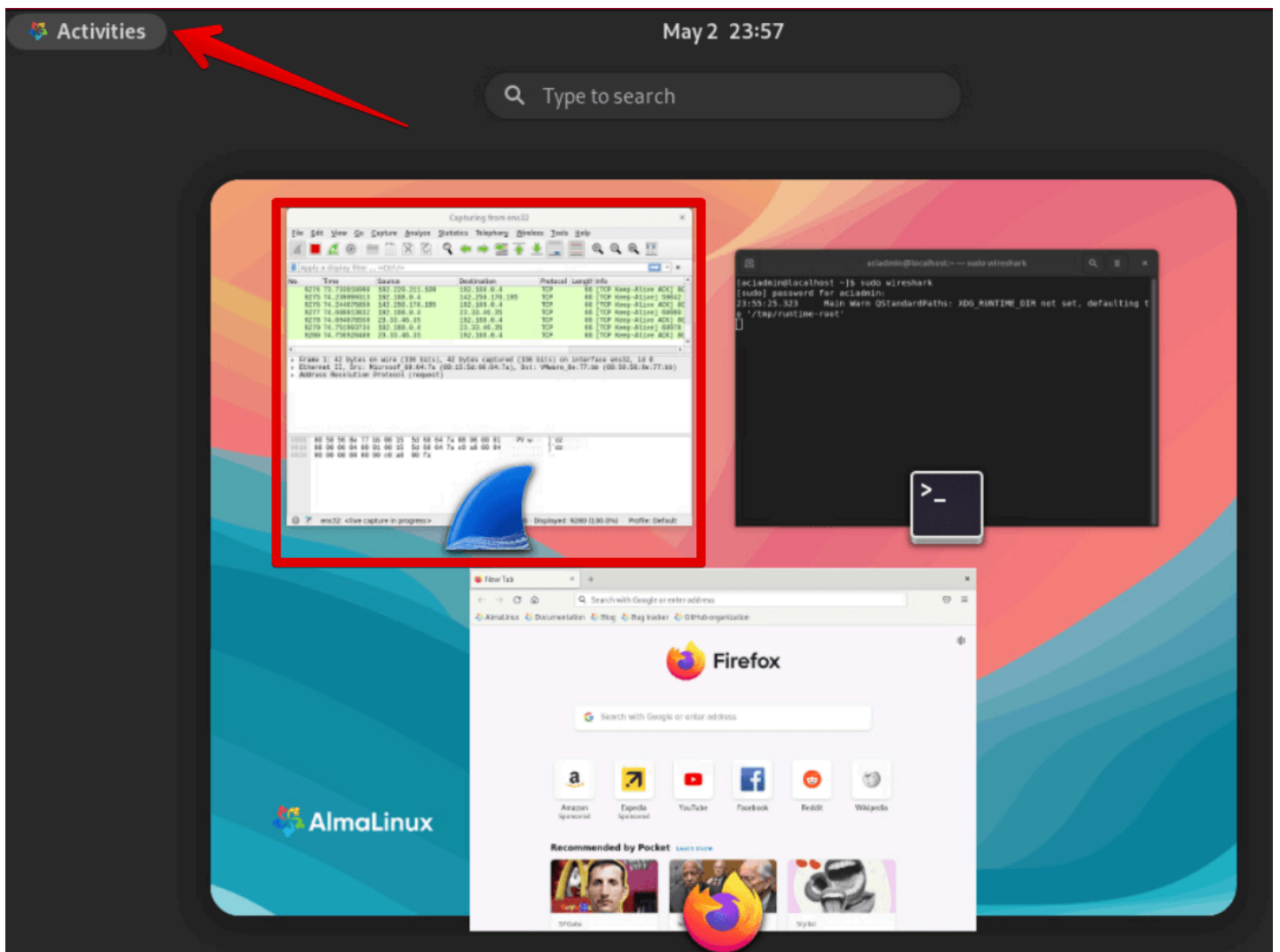
Click **Activities** and **select Firefox**.



Note: Opening the **Firefox browser** will generate web traffic captured by the Wireshark application.

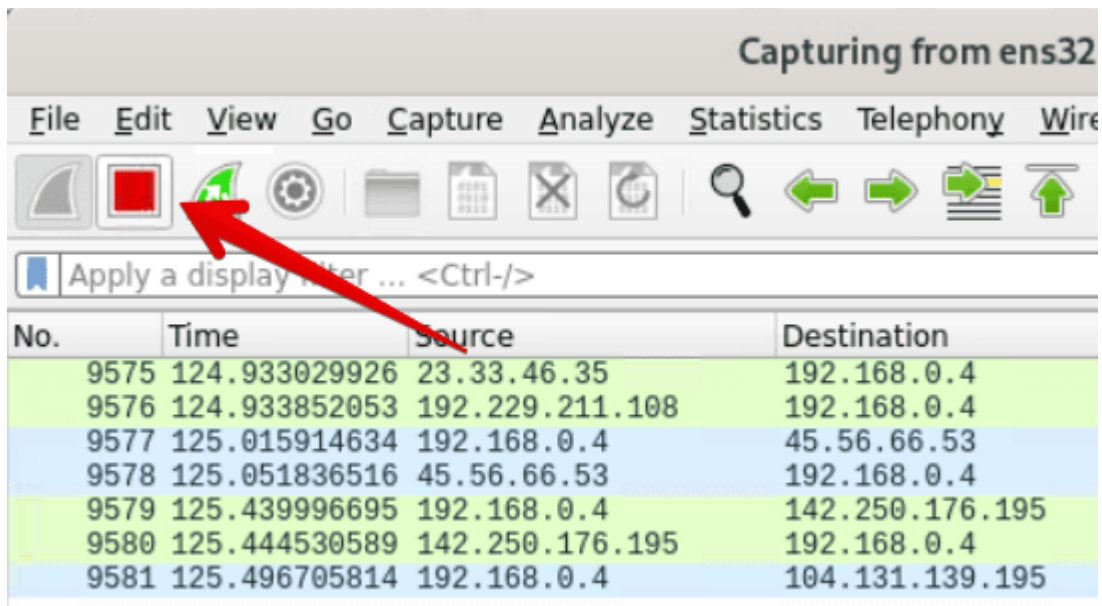
6

Click **Activities** and **select the Wireshark application**.



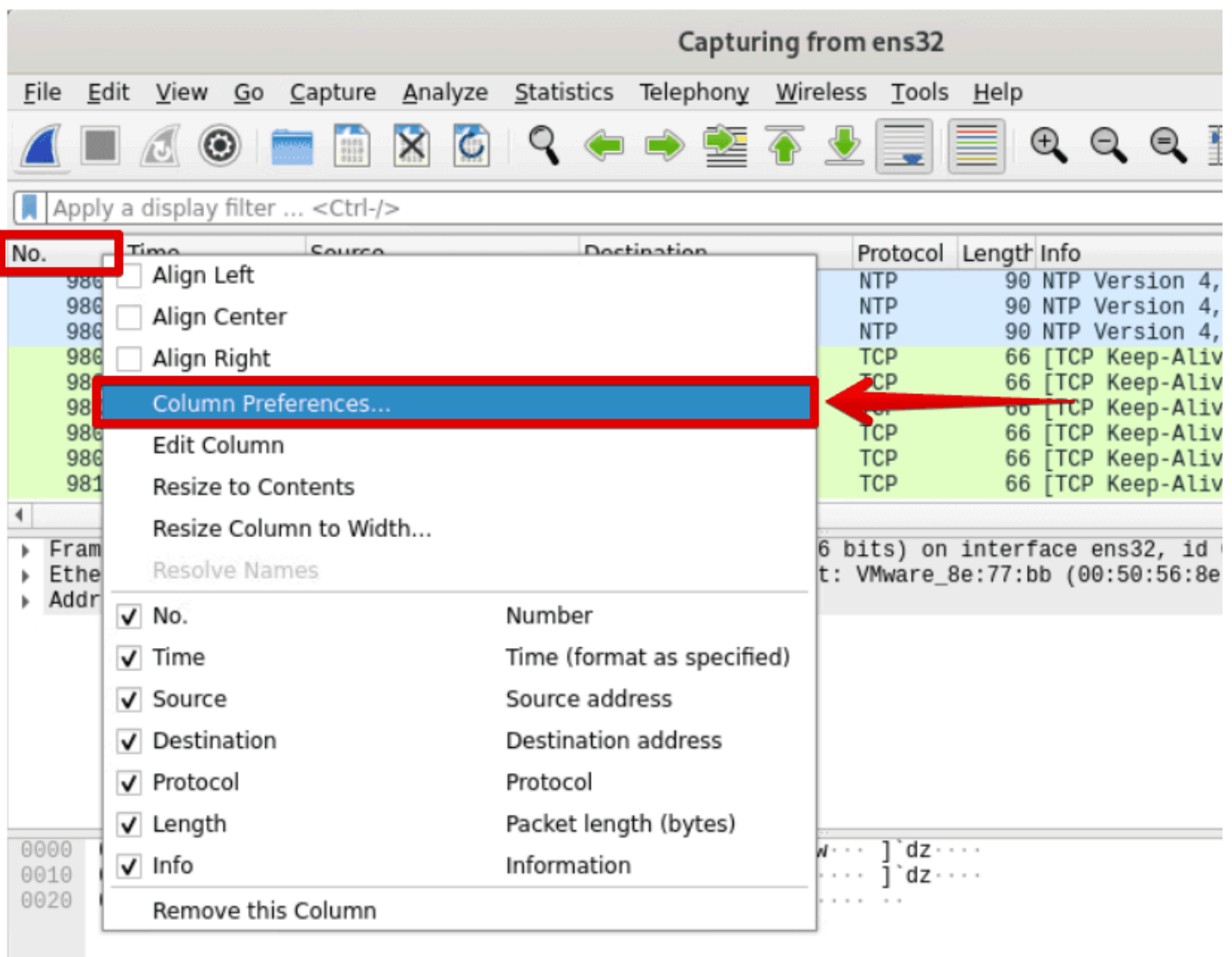
7

In the **Wireshark** application, **click** Stop capturing packets.



8

In the **Wireshark** application, **right-click** the **No. Column** and **select** Column Preferences.



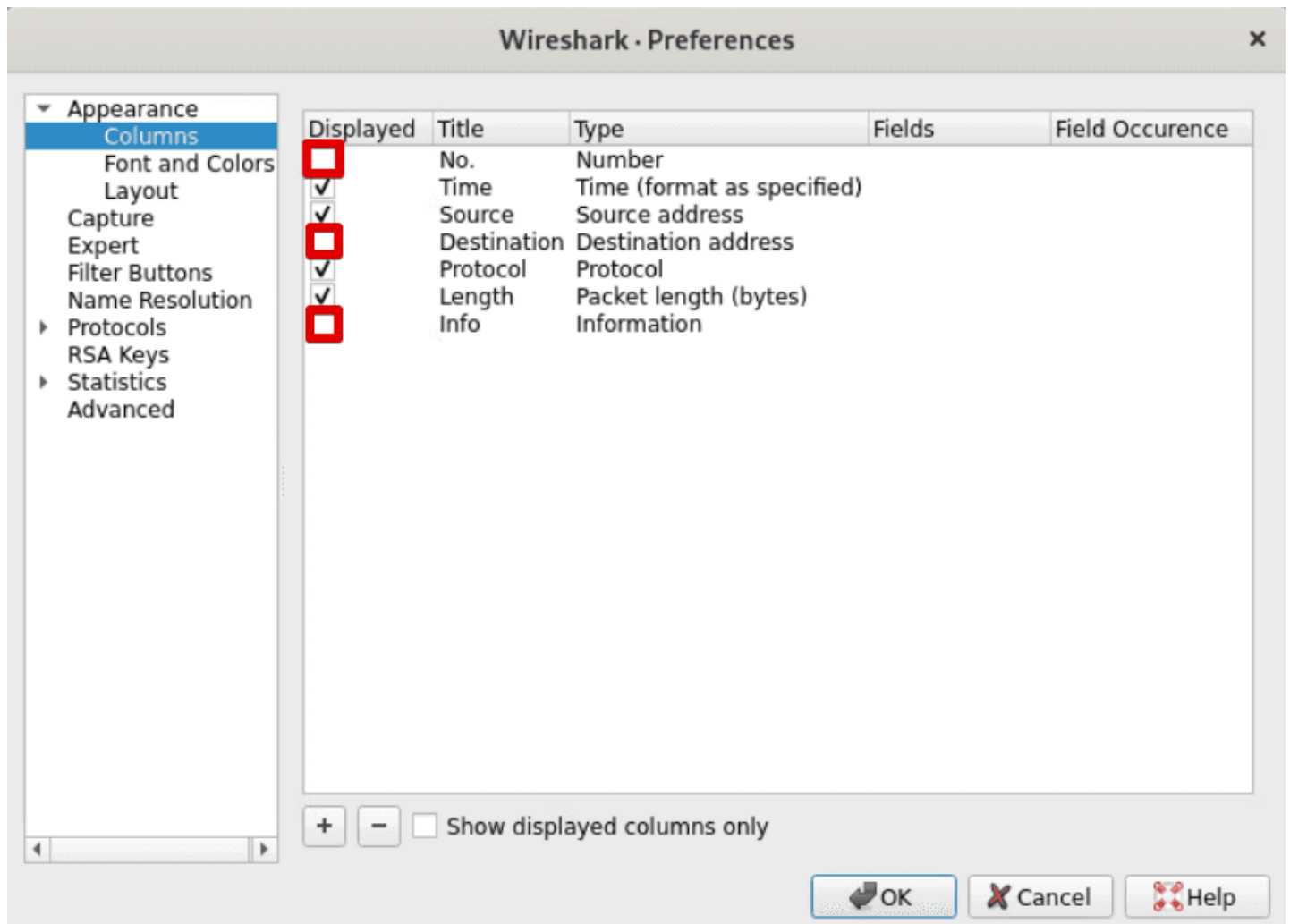
9

In the **Wireshark - Preferences** pop-up window, **untick** the following tick-boxes:

No.

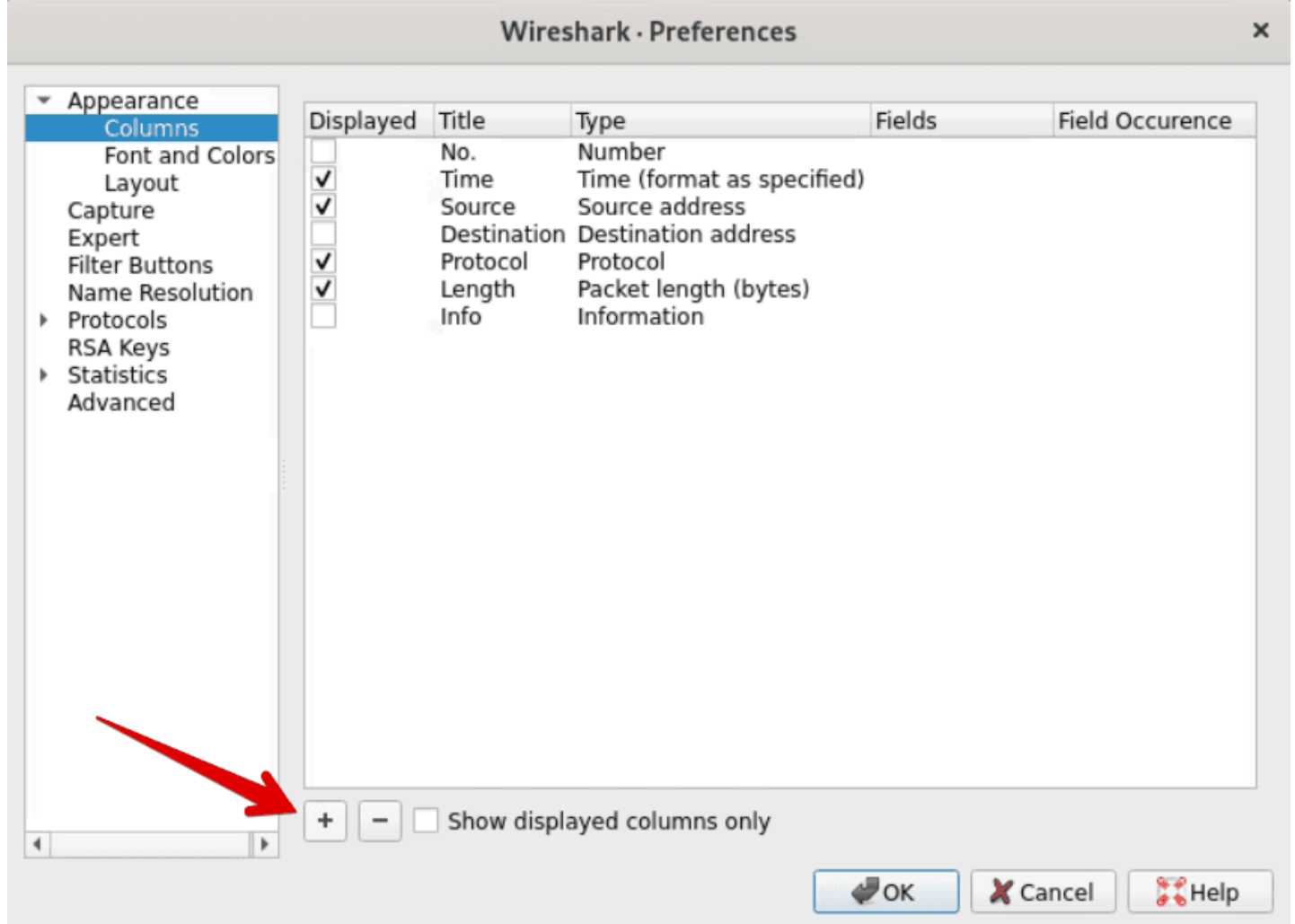
Destination

Info



10

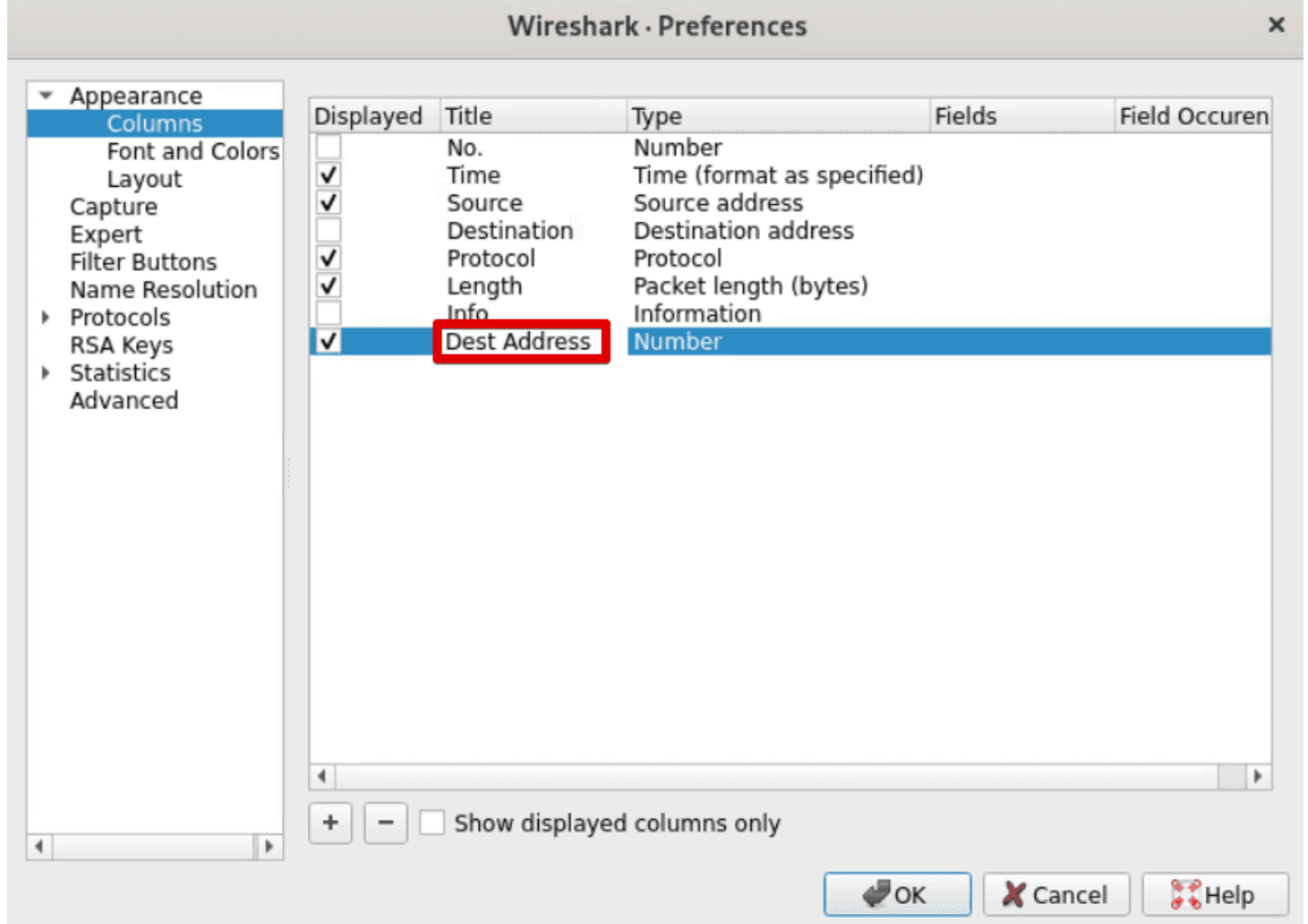
Click the **+** Add a new column button at the bottom of the screen.



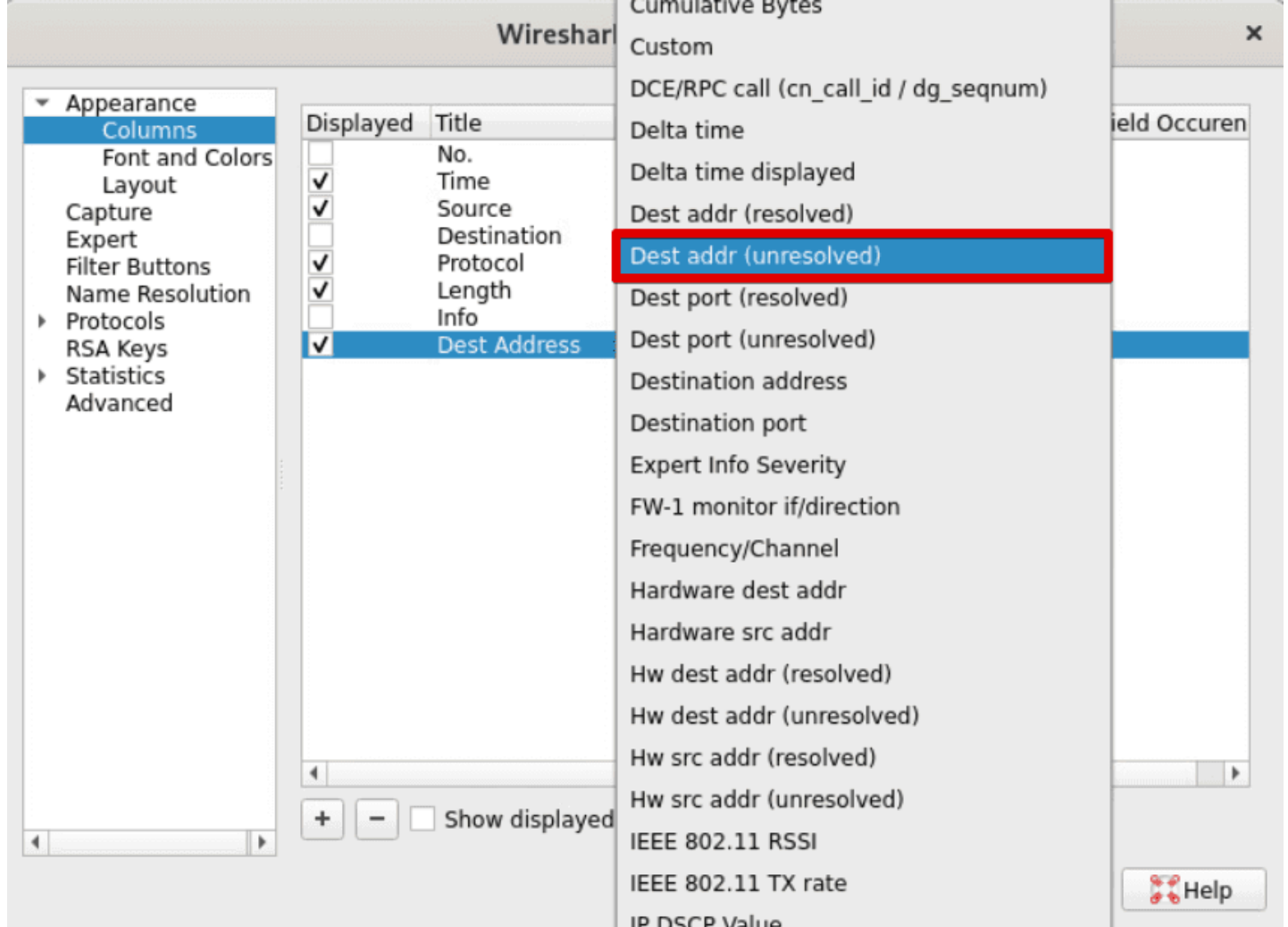
11

Double-click New Column in the Title column and **type** the following:

Dest Address

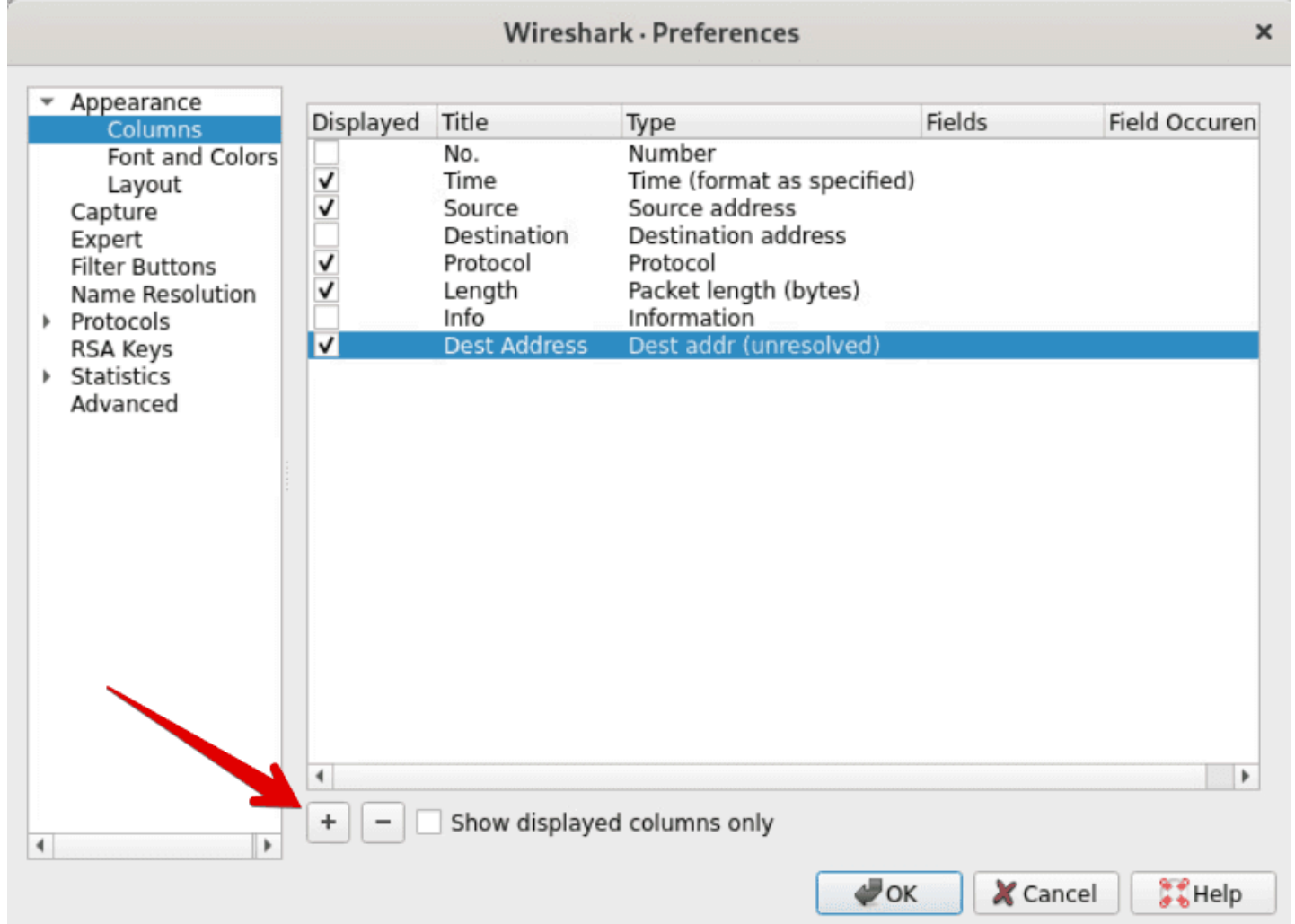


12 **Double-click** Number in the Type column and **select** Dest addr (unresolved) in the drop-down menu.



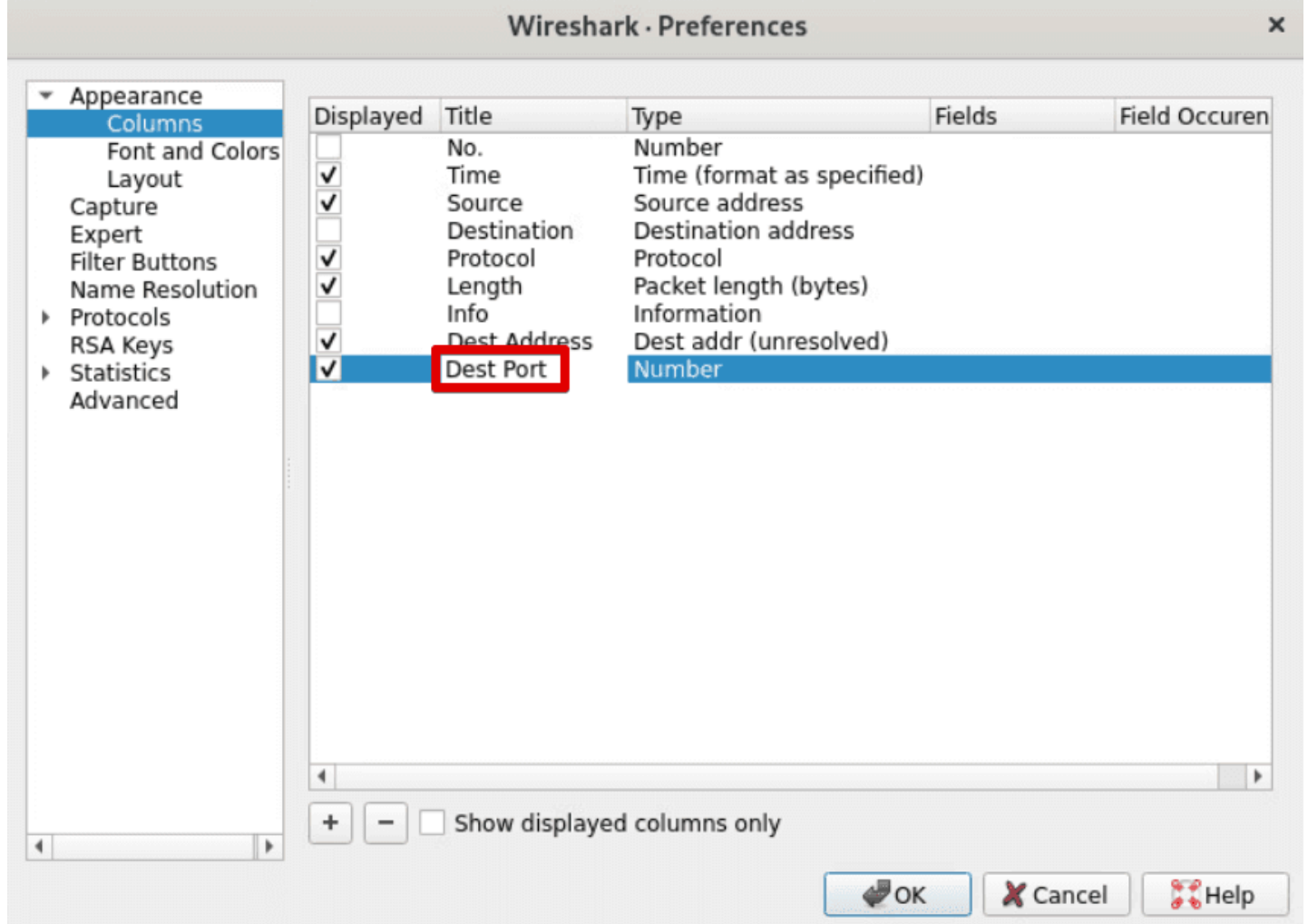
13

Click the + **Add new column** at the bottom of the screen.

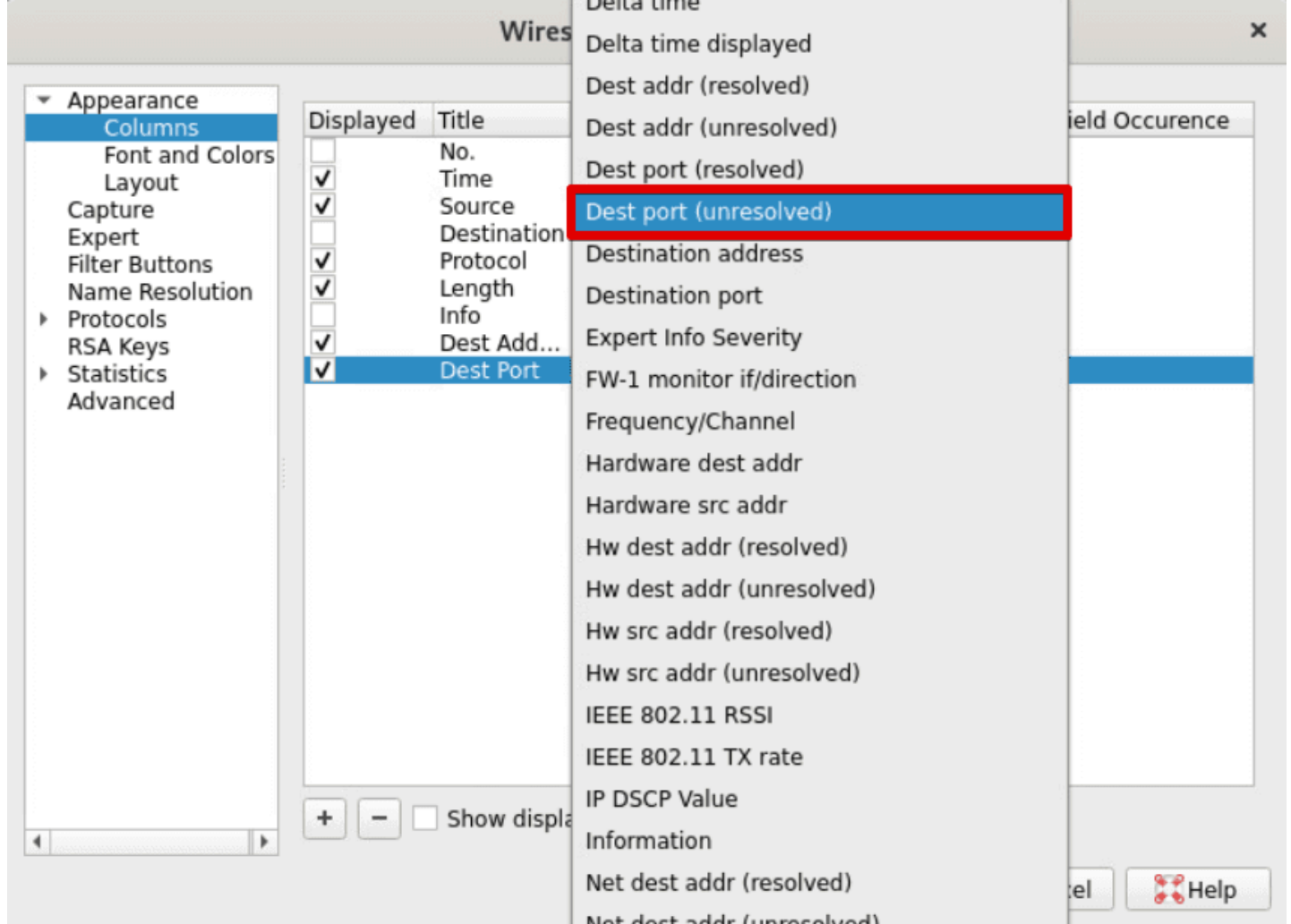


14 In the **Title** column, **double-click** **New Column** **enter** the following:

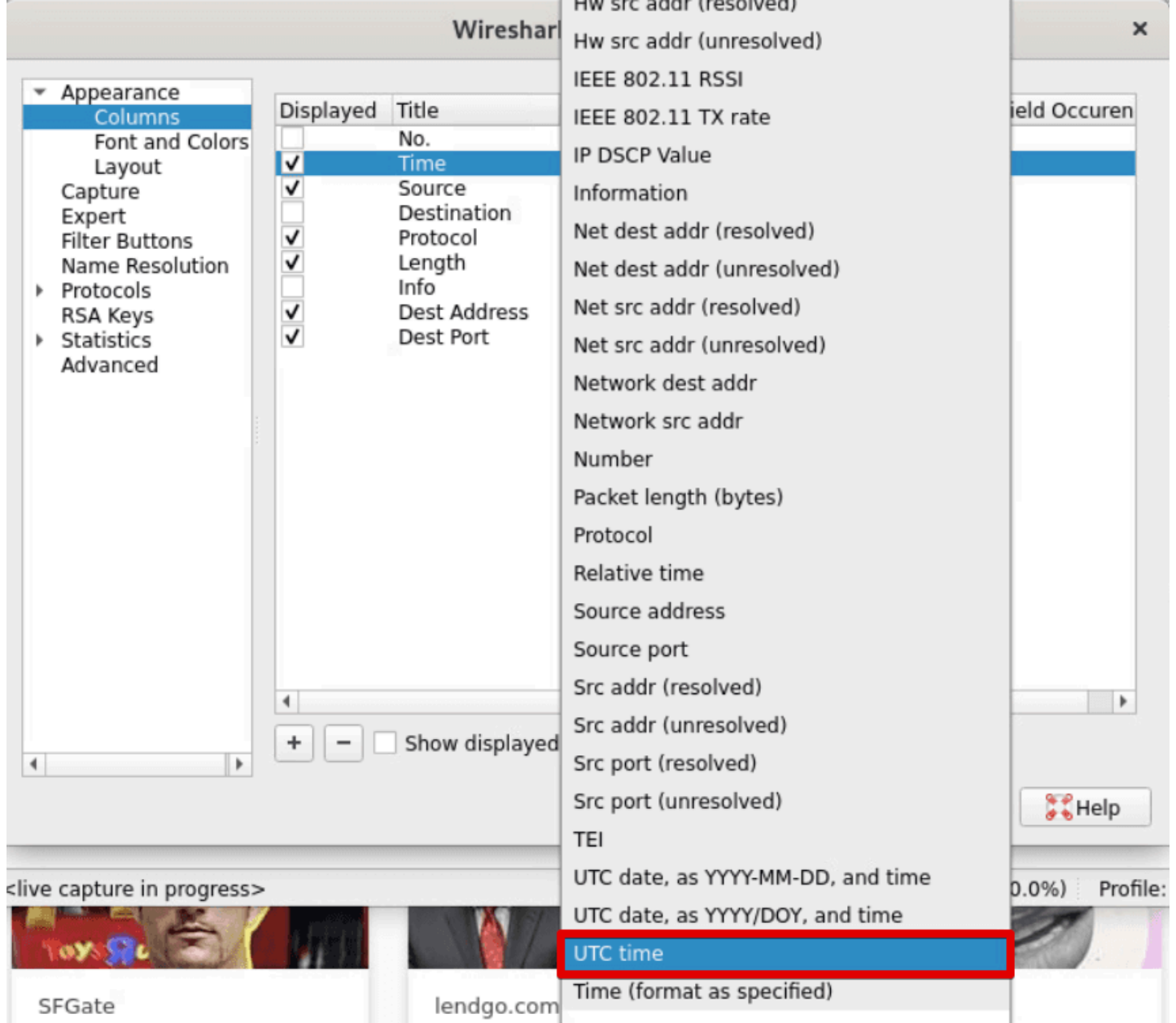
Dest Port



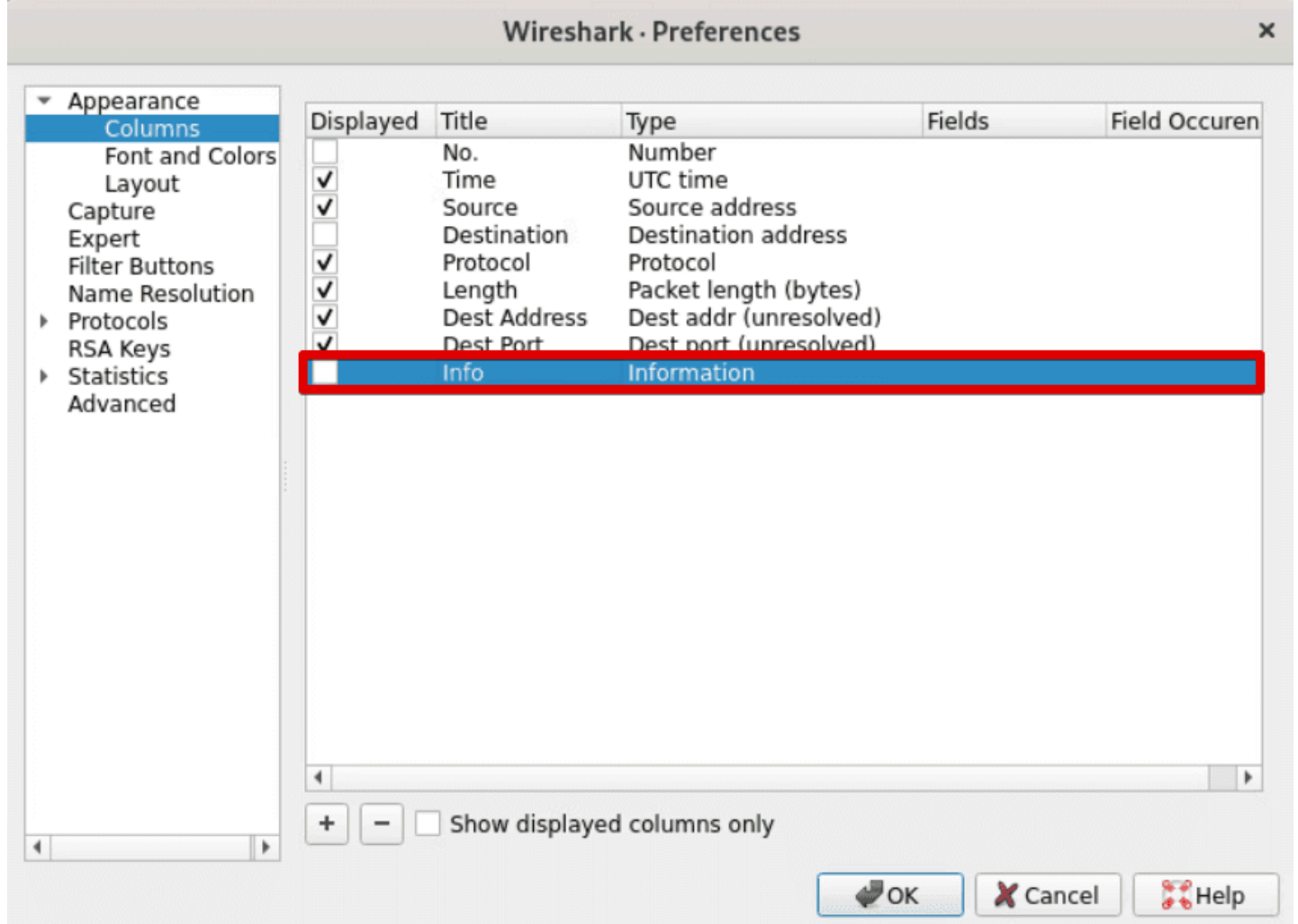
- 15 **Double-click** Number in the Type column and **select** Dest port (unresolved) in the drop-down menu.



16 Double-click Time (Format as specified) in the Type column and select UTC Time.

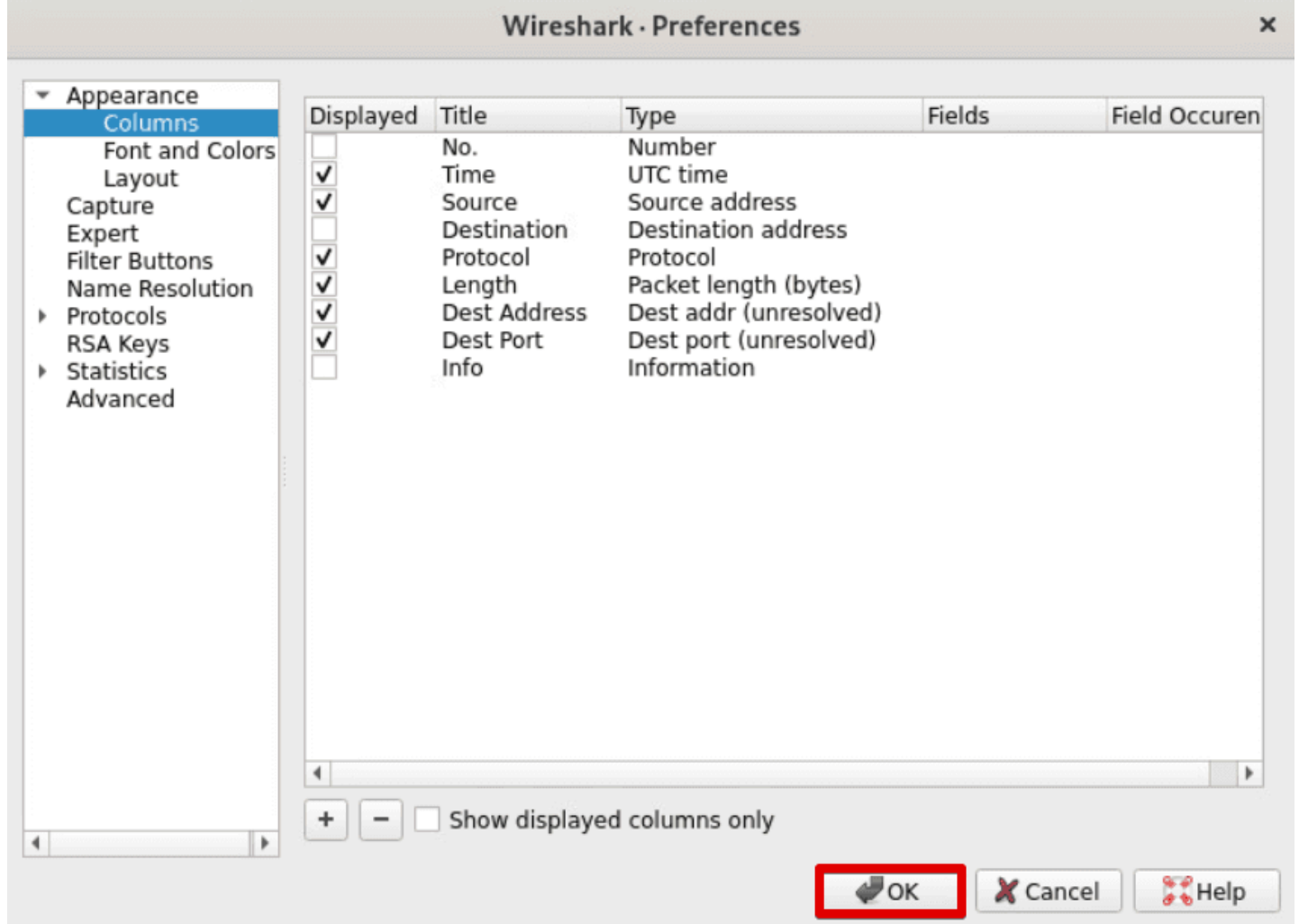


- 17 In the **Wireshark - Preferences** window, **select Info** and **drag** and **drop it** below **Dest Port**.



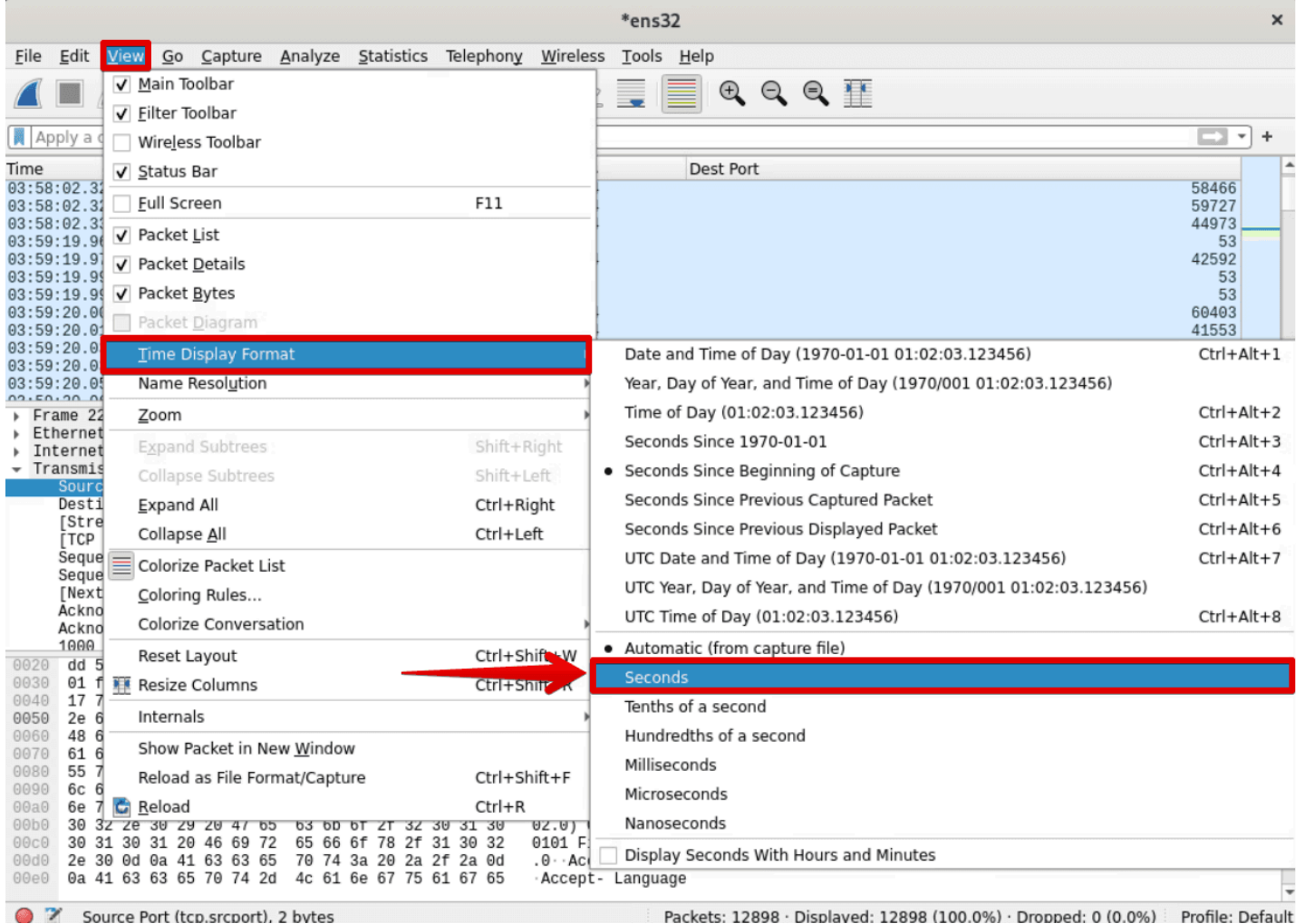
18

Click **OK** on the **Wireshark - Preferences** pop-up window.



19

Click the **View** menu, **select** Time Display Format and **click** Seconds.



Note: Leave the Wireshark application open for the next task.



Note: The columns can be added or removed depending on the network traffic type that Cyber Security Analyst needs to monitor and analyze.

Task 2 – Capturing Network Traffic using the Wireshark Application

In this task, the Wireshark application will be used to capture network traffic

- 1 In the **Apply a display filter**, **enter** the following and **click** **Apply display filter**:

http

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

http

Time	Source	Protocol	Length	Dest Address	Dest Port
04:12:43	192.168.0.3	SSDP	179	239.255.255.250	1900
04:12:46	192.168.0.3	SSDP	179	239.255.255.250	1900
04:12:52	192.168.0.4	TCP	66	23.33.46.35	80
04:12:52	23.33.46.35	TCP	66	192.168.0.4	35552
04:13:03	192.168.0.4	TCP	66	23.33.46.35	80
04:13:03	23.33.46.35	TCP	66	192.168.0.4	35552
04:13:12	192.168.0.4	NTP	90	45.56.66.53	123
04:13:12	45.56.66.53	NTP	90	192.168.0.4	41804
04:13:13	192.168.0.4	TCP	66	23.33.46.35	80
04:13:13	23.33.46.35	TCP	66	192.168.0.4	35552

Frame 1: 42 bytes on wire (336 bits), 42 bytes captured (336 bits) on interface ens32, id 0
 Ethernet II, Src: Microsof_60:64:7a (00:15:5d:60:64:7a), Dst: VMware_8e:77:bb (00:50:56:8e:77:bb)
 Address Resolution Protocol (request)

0000 00 50 56 8e 77 bb 00 15 5d 60 64 7a 08 06 00 01 ·PV·w··· }`dz····
 0010 08 00 06 04 00 01 00 15 5d 60 64 7a c0 a8 00 04 ······ }`dz····
 0020 00 00 00 00 00 00 c0 a8 00 fa ······

Hypertext Transfer Protocol: Protocol Packets: 12829 · Displayed: 12829 (100.0%) Profile: Default



Note: A filter has been applied to display specific network traffic that is being captured. In this example, the http traffic is being filtered.

2

Select the **first record** on the captured packages and **expand** **Transmission Control Protocol** in the middle pane.

3

Click **Clear display filter** in the **Wireshark** application.

4

In the **filter field**, **enter** the **following**:

```
ip.addr == 192.168.0.0/24
```

Click **Apply display filter**.

5

In the **Wireshark application**, **click** the **Source address column**.



Note: Applying the **ip. addr == 192.168.0.0/24** filter displays all the traffic from the different hosts captured by the Wireshark application. A Cyber Security Specialist can use this information to determine the different types of traffic from different hosts and if it is legitimate traffic.

6

In the **Wireshark application**, **select** the **Protocol column**.



Note: The captured traffic can be filtered by clicking the specific columns according to the Port, Destination or Source IP addresses.

7

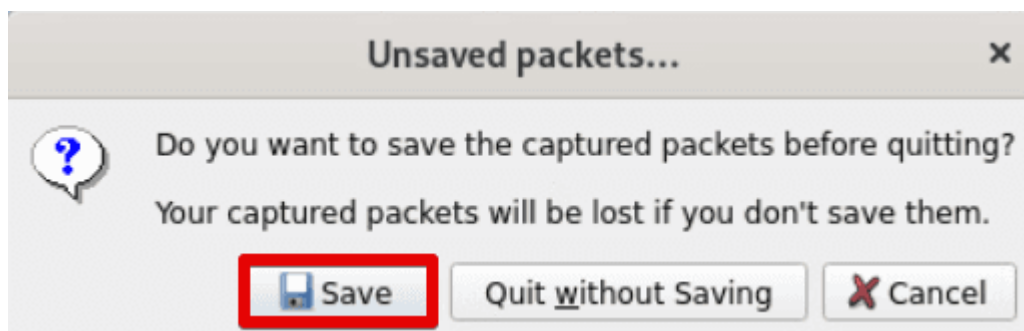
Close the **Wireshark application**.



Note: Depending on the traffic a Cyber Security Analyst needs to monitor, different filters can be applied to the Wireshark application. Please visit the following website to investigate a full list of different types of filters that can be applied:
<https://www.wireshark.org/docs/man-pages/wireshark-filter.html>

8

In the **Unsaved packets pop-up window**, **click** **Save**.



9

In the **Wireshark -Save Capture As**, **type** the **following** in the **File Name field** and **click**

Save:

ACI Network Traffic



Note: The captured packets can be saved, which can be analyzed later or used as part of an investigation by a Cyber Security Analyst.

Task 3 – Capture Network Traffic using Tcpdump

The tcpdump application is a command line utility that can be used to capture network traffic.

In this task, the tcpdump command line utility will be used to capture network traffic.

1

Connect to **ACIALMA**, **click** **Activities** and **select** the **Terminal window**.

2

In the **Terminal window**, **type** the **following**:

```
sudo tcpdump --list-interfaces
```

Press **Enter**.



Note: The command is executed to determine which network interface on the device will be used to capture network traffic.

3

Connect to **ACIWIN11**.

Click **Send Ctrl+Alt+Delete** button on the top of the screen.

Username: **ACIPLAB\Administrator**

Password: **Passw0rd**

4

Right-click **Start** and **select** **Windows Terminal (Admin)**.

5

In the **Windows Terminal window**, **type** the following:

```
ping 192.168.0.4 -t
```

6

Connect to **ACIALMA** and the following in the **Terminal window**:

```
sudo tcpdump -n icmp -c 10 -vv -w icmptraffic.pcap
```

Press Enter.



Note: The tcpdump command can be executed with different parameters to capture network traffic. In this example, it is used to capture ICMP traffic generated from the ACIWIN11 machine and saved to a .pcap file.

7

In the **Terminal window**, **type** the following:

```
sudo wireshark icmptraffic.pcap
```

Press Enter.



Note: The captured traffic using the tcpdump command was saved to a file and can be viewed using the Wireshark application.

8

Close the **Wireshark application**.

9

In the **Terminal window**, **type** the following:

```
clear
```

Press Enter.

```
sudo tcpdump -i ens32 -nn -v port 80 -w httptraffic.pcap
```

Press Enter.



Note: The tcpdump utility can be used to capture various network traffic using different parameters. To see more details of the different parameters, type the following command:

```
man tcpdump
```

10

Click **Activities** and **select** **Firefox**.

11

Click **Activities** and **select** the **Terminal window**.

12

In the **Terminal window**, **press** **CTRL +C**.

13

In the **Terminal window**, **type** the **following**:

```
sudo wireshark icmptraffic.pcap
```

Press **Enter**.

14

In the **Wireshark application**, **select** the **Dest Address column**.



Note: The captured traffic can be filtered accordingly using the Wireshark application.

15

Connect to **ACIWIN11** and **ensure** the **Windows Terminal** is open.

Press **CTRL + C**.

Exercise 2 – Using a Sandbox for Analyzing Malicious Files

A Sandbox solution can be used to analyze potentially harmful or malicious files. Windows 11 has a sandbox solution that can execute these malicious or harmful files without affecting the host operating system.

In this exercise, a malicious will be executed in Windows Sandbox.

Learning Outcomes

After completing this exercise, you should be able to:

- Analyze Malicious Code

Your Devices

You will be using the following devices in this lab. Please power these on now.

- **ACIDC01** – Windows Server 2022 – Domain Controller
- **ACIWIN11** – Windows 11 – Domain Member Workstation

Task 1 – Malicious Code Analysis

In this task, a potentially malicious file will be executed in a sandbox solution.

1

Connect to **ACIWIN11**.

Click the **Send Ctrl+Alt+Delete button** on the top of the screen.

Sign-in with the following credentials:

Username: ACIPLAB\Administrator

Password:

Click Start, and **select** All apps.

2

Scroll down on the menu and **select** Windows Sandbox.

3

From the Taskbar of the ACIWIN11 device, **select** File Explorer.

4

In File Explorer, **select** Documents, **right-click** Good bye and **select** Copy.

5

Open Windows Sandbox from the Taskbar.

6 **Right-click** on the **Desktop** of the **Windows Sandbox** device and **select Paste**.

7 **Double-click** the **Good bye file**.



Note: The Windows Sandbox is an additional feature of Windows 11. A Cyber Security Analyst can utilize this feature to execute suspected malicious code in an isolated environment. In this case, a suspected batch file is copied into the Sandbox to verify what will happen if a user executes the file.

8 **Click OK** on the **pop-up window**.

9 **Click OK** on the **Windows Sandbox pop-up window**.



Note: After the file has been executed, the device will be restarted. This indicates that the file is malicious and could potentially cause damage if executed by the user.

10 **Click Start**, **select All apps** and **open Windows Sandbox** from the menu.

11 **Close** the **Windows Sandbox window**.



Note: Reopening the Windows Sandbox shows that any changes made to the device have been reverted. A Cyber Security Analyst can use the application to test the impact an application will have on the device without having any persistent changes.

12

Click **OK** on the **Windows Sandbox** pop-up window.

Close all **open windows**.

Exercise 3 – Validating Domain Names and IP Addresses

The WHOIS website can be used to resolve IP addresses to Domain Names. As a Cyber Security Specialist, it will be needed to validate the IP addresses that access resources on the network. In addition to validating IP addresses, the WHOIS website can be used to validate domain names.

In this exercise, the WHOIS website will resolve IP addresses to Domain Names and validate domain names.

Learning Outcomes

After completing this exercise, you should be able to:

- Use WHOIS to Validate Domain Names
- Use AbuseIPDB to Validate Domain Names
- Use the Virustotal Website to Verify Links

Your Devices

You will be using the following devices in this lab. Please power these on now.

- **ACIDC01** – Windows Server 2022 – Domain Controller
- **ACIWIN11** – Windows 11 – Domain Member Workstation

Task 1 – Use WHOIS to Validate Domain Names

In this task, the WHOIS website will be used to validate IP addresses and Domain Names.

1

Connect to ACIWIN11.

Open Microsoft Edge from the Taskbar.

2

In the Microsoft Edge browser, **Type** the following URL:

`https://www.whois.com/`

3

In the WHOIS search bar, **type** the following in the Enter Domain or IP textbox and

press Enter:

`13.107.42.14`



Note: The Whois website has resolved the IP address to a Domain Name. From the results, it can be seen that Microsoft owns the IP address. A Cyber Security Analyst can enter detected IP addresses to determine the source or owner.

4

In the **WHOIS** search bar, **enter** the **following** and **press Enter**:

`www.acilearning.com`



Note: Information on the acilearning Domain Name is displayed. From the results, it can be seen that the website is hosted on Amazon Web Services. There is also further information on the Domain Name.

Task 2 – Use AbuseIPDB to Validate Domain Names

The AbuseIPDB website can verify if an IP address has been reported for abuse.

In this task, the AbuseIPDB website will validate an IP address and Domain Name.

1

Browse the **following website** in the **Microsoft Edge browser**:

`https://www.abuseipdb.com`

2

In the **CHECK** field, **enter** the **following** and **press Enter**:

`comptia.org`



Note: The website's IP address is resolved from the Domain Name. From the results, it can be seen that it has not been reported for abuse.

3

Click on **WHOIS 172.64.152.227** on the results page.



Note: The IP address displayed will differ when conducting these steps.



Note: The AbuseIPDB uses the Whois website to resolve the IP address to a Domain Name, which is resolved to comptia.org. More information about the website, for example, the different subdomains, can be found.

4

Enter the following IP address in the CHECK field:

172.64.152.227

5

Close the Microsoft Edge browser.



Note: The entered IP address can be seen and has been reported several times, and the location is in South Africa. Scrolling down the results page, the different reports can be seen.

Task 3 – Use the Virustotal Website to Verify Links

The Virustotal website can be used to verify Domain Names, IP addresses, or suspicious links received via email.

In this task, the Virustotal website will be used to investigate a suspicious IP address.

1

Connect to ACIWIN11, **open** Microsoft Edge from the Taskbar and **browse** to the following URL:

<https://virustotal.com/>

2

Click Search on the virustotal website.

3

In the search field, **enter** the following:

196.242.35.45

Press Enter.

4

Click the Back arrow in the Microsoft Edge browser.



Note: The Virustotal website only recorded one abuse report on the specified website. The results may vary from different vendors depending on the number of reports for a specific IP address.

5

Select URL on the **virustotal** website.

6

Enter the **following** in the **Search** or **scan a URL** field:

`www.practice-labs.com`

Press **Enter**.

7

Click **Details** on the **results** page.

8

Close the **Microsoft Edge** browser after viewing the **DETAILS** tab.



Note: More on the URL information is available and can be used to determine if the website has potential risks. The Virustotal website can also be used to upload links received via email, which can then be analyzed to determine if it is malicious.



Keep all devices that you have powered on in their current state and proceed to the review section.

Review

Well done, you have completed the **Tools for Identifying Malicious Activity** practice lab.

Summary

You completed the following exercises:

- Exercise 1 – Monitoring Network Activity
- Exercise 2 – Using a Sandbox for Analyzing Malicious Files
- Exercise 3 – Validating Domain Names and IP Addresses

You should now be able to:

- Customize the Wireshark Application
- Capture Network Traffic using the Wireshark Application
- Capture Network Traffic using Tcpdump
- Analyze Malicious Code
- Use WHOIS to Validate Domain Names
- Use AbuseIPDB to Validate Domain Names
- Use the Virustotal Website to Verify Links

Feedback



Shutdown all virtual machines used in this lab. Alternatively, you can log out of the lab platform.